

Change Detection & Image Time Series Analysis

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Outline

Change Detection

- Introduction

- Requirements

- Post-classification

- Change Thresholds

- Binary Masking

Image Time Series Analysis

- Introduction

- Landscape Phenology

- A Greenness Profile

- Future Directions

Plan

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What is Change Detection?

- ▶ **Definition:** The process of identifying and characterizing differences in conditions at different points in time.
- ▶ **Value:** One of the most powerful advantages of remote sensing is preserving a temporal record.
- ▶ **Scales of Change:**
 - ▶ *Short-term:* Vehicle or animal movement.
 - ▶ *Long-term:* Urban development or desertification.

Technical Requirements

To minimize “false” changes, data should ideally be acquired using:

- ▶ Same or similar **sensors**.
- ▶ Identical **spatial resolution** and viewing geometry.
- ▶ Matching **spectral bands** and radiometric resolution.
- ▶ Consistent **time of day** to maintain sun angle.
- ▶ **Anniversary dates**: Preferred for long-term (>1 year) studies to minimize seasonal differences.

Registration and Environmental Factors

Spatial Accuracy

Effective detection requires accurate registration, ideally within $\frac{1}{4}$ to $\frac{1}{2}$ pixel. Misregistration > 1 pixel causes significant errors.

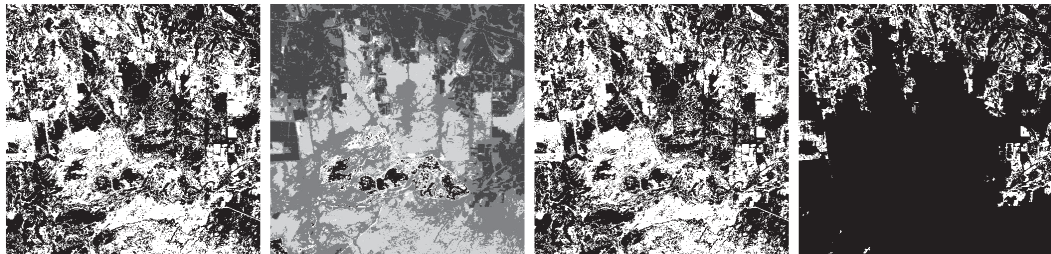
► **Environmental Influences:**

- Atmospheric effects.
- Water levels and tidal stages.
- Soil moisture and wind.
- Plant phenology (season-to-season biological changes).

Method 1: Post-classification Comparison

- ▶ **Process:** Two dates are independently classified and registered.
- ▶ **Mechanism:** Pixels that changed class (e.g., Class A $\rightarrow$ Class B) are identified.
- ▶ **Pros:** Provides specific “from-to” change maps and statistics.
- ▶ **Cons:** Accuracy depends on the precision of both independent classifications; **errors are compounded.**

Post-classification change detection



(a) classification on date 1; (b) classification on date 2; (c) all change pixels highlighted in white; (d) only pixels that changed from class “shrubland” to class “agriculture”.

Method 2: Classification of Multi-temporal Data

- ▶ **The “Stacking” Approach:** Spectral bands from Date 1 and Date 2 are combined into a single data set (e.g., two 4-band images become one 8-band image).
- ▶ **Analysis:** Supervised or unsupervised classification is run on the stack.
- ▶ **Classes:** Resulting classes represent either stable land cover or specific “change classes”.
- ▶ **Challenge:** High dimensionality and potential redundancy in information.

Method 3: Principal Components Analysis

- ▶ **Approach:** Multiple dates are registered into a multiband image, and PCA is computed.
- ▶ **Change Detection:** Uncorrelated principal components often highlight areas of change (e.g., detecting tornado damage paths).
- ▶ **Advantages:** Excellent for identifying distinct physical changes.
- ▶ **Disadvantages:** It is often difficult to interpret the specific nature of the change.

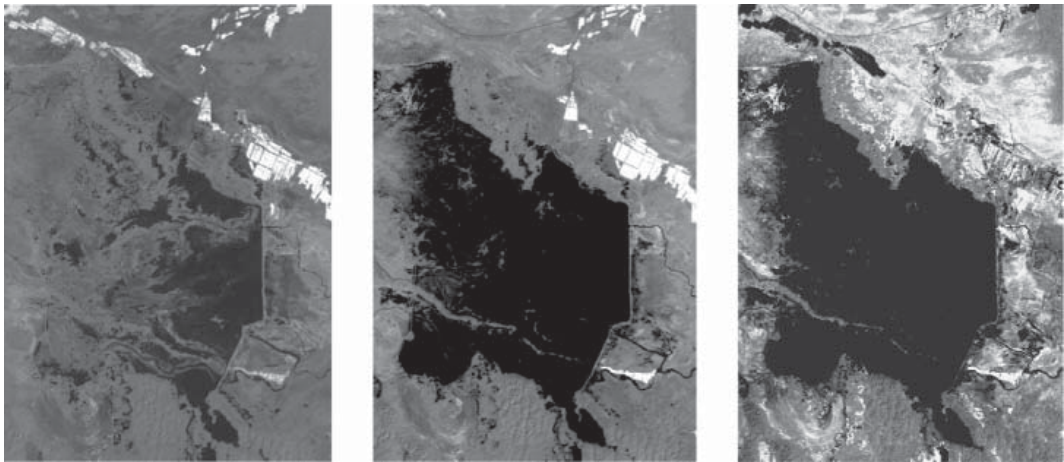
Method 4: Temporal Image Differencing

- ▶ **Calculation:** Digital Numbers (DNs) from Date 1 are subtracted from Date 2.
- ▶ **Range:** For 8-bit images, results range from -255 to $+255$.
- ▶ **Interpretation:**
 - ▶ *Near Zero:* No change (mid-gray tone).
 - ▶ *Positive/Negative:* Large changes (bright or dark tones).
- ▶ **Display:** A constant (e.g., 127 or 255) is often added for visualization.

Method 5: Temporal Image Ratioing

- ▶ **Calculation:** Computing the ratio of data between two dates.
- ▶ **Logic:** Areas of no change result in a ratio of 1.
- ▶ **Advantage:** Tends to normalize data for extraneous factors like **sun angle** and **shadows**.
- ▶ Like differencing, results are typically scaled for display.

Temporal image differencing



(a) near-IR band from date 1; (b) near-IR band from date 2; (c) difference of (b) minus (a).

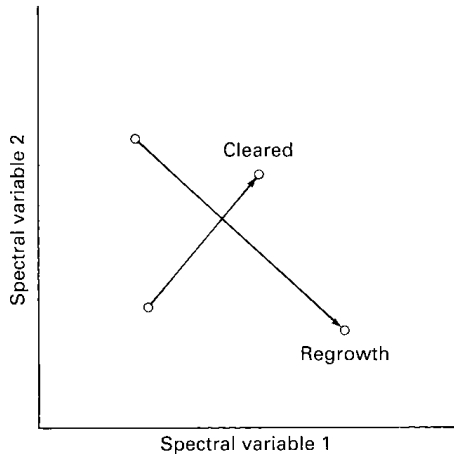
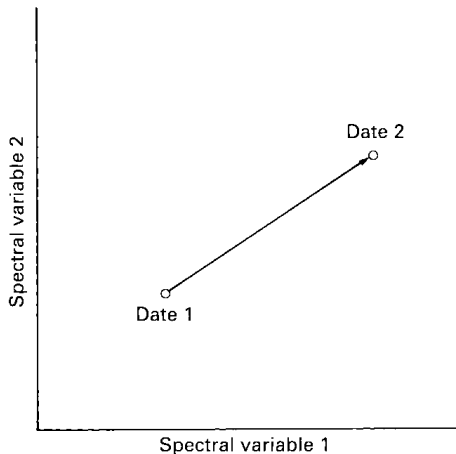
Determining “Change” Thresholds

- ▶ **The Threshold:** Analysts must find a “change-no change” threshold by looking at the **histogram tails** of differenced or ratioed data.
- ▶ **Enhancements:** Rather than raw DNs, analysts often use:
 - ▶ Atmospheric/Illumination corrections.
 - ▶ Radiance or Reflectance values.
 - ▶ Vegetation components or Linear Regression models.

Method 6: Change Vector Analysis

- ▶ **Concept:** A conceptual extension of differencing using two or more spectral variables.
- ▶ **The Vector:** Describes both **magnitude** and **direction** of spectral change.
- ▶ **Applications:**
 - ▶ Magnitude threshold determines if a change occurred.
 - ▶ Direction identifies the *type* of change (e.g., clearing vs. regrowth).

Spectral change vector analysis



(a) spectral change vector observed for a single land cover type; (b) length and direction of spectral change vectors for hypothetical “cleared” and “regrowth” areas.

Efficiency and Automation

- ▶ **Binary Masking:** Create a change-vs-no-change mask (via differencing/ratioing) to focus classification only on areas where change occurred.
- ▶ **Conclusion:** Change detection is increasingly vital due to the massive volume of data from:
 - ▶ Expanded satellite constellations.
 - ▶ Airborne platforms.
 - ▶ Rapidly expanding UAV (drone) technology.
- ▶ **Future:** Emphasis on **automated** methods to interpret the significance of changes.

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The Rise of Time Series Analysis

- ▶ **Beyond Snapshots:** Traditional change detection uses pairs of images; Time Series Analysis uses hundreds.
- ▶ **Global Monitoring Systems:** Enabled by sensors like **AVHRR** and **MODIS**.
- ▶ **Data Richness:** Daily to monthly intervals over decades.
- ▶ **Purpose:** Captures seasonal to decadal variations in:
 - ▶ Agricultural cycles.
 - ▶ Ecological health.
 - ▶ Hydrological systems.

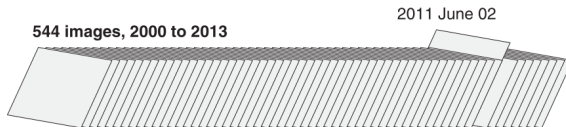
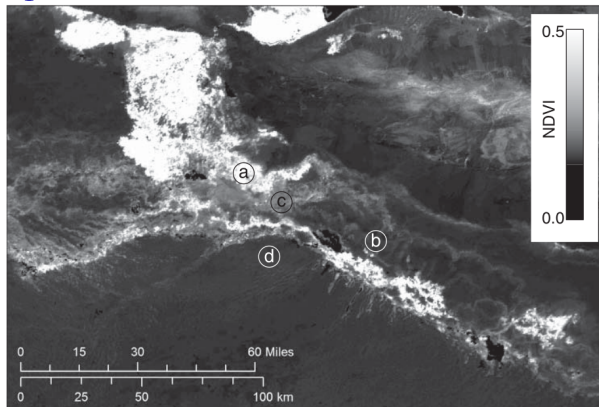
What is a Temporal Signature?

- ▶ Every pixel is a “link in a long chain” of imagery (e.g., 544 consecutive MODIS images).
- ▶ **Extraction:** By tracking a single pixel over time, we create a **temporal signature**.

Case Study: Xinjiang Region

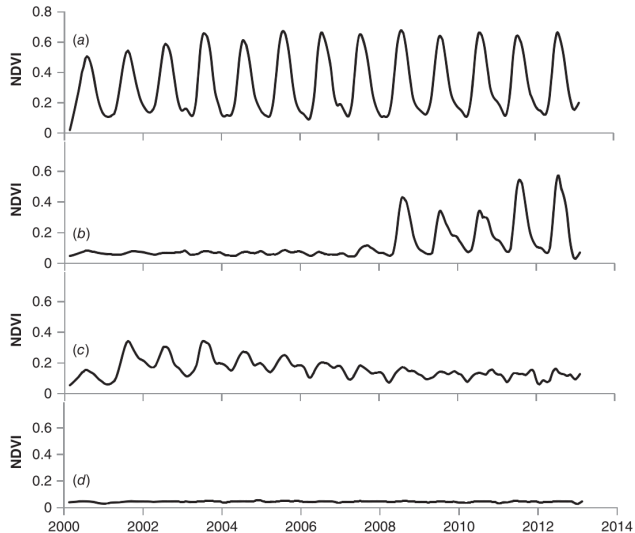
- ▶ **Agriculture:** Sharp cyclical peaks (planting to harvest).
- ▶ **Desert:** A flat, near-zero line.
- ▶ **Flooded Shrubland:** Sudden spikes in greenness followed by gradual decay.

Time Series Example I



MODIS NDVI imagery extracted from a time series of 544 images for the Tarim River area in Xinjiang, China, from February 2000 through February 2013. Labeled points (a-d) are sources for time-series graphs shown in next figure.

Time Series Example II



Time-series plots of previous MODIS NDVI figure: (a) established cotton farms; (b) new cotton farms; (c) tamarisk shrubland following an episode of flooding; (d) sand dunes.

Landscape Phenology

- ▶ **Definition:** The study of spatial-temporal variations in vegetation indices (like NDVI or EVI) across the landscape.
- ▶ **Key Research Areas:**
 - ▶ Linkages between climate and ecosystems.
 - ▶ Agricultural yield forecasting.
 - ▶ Hydrologic monitoring and snow/ice cover changes.

“It is the pulse of the annual cycle of green-up and senescence”.

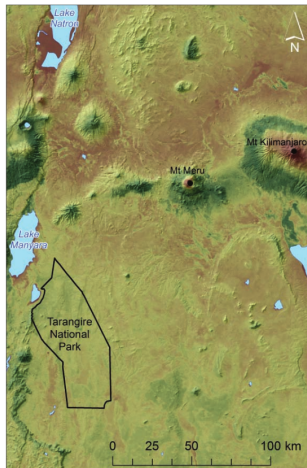
Case Study: Northern Tanzania (2000–2011)

Using *the Enhanced Vegetation Index (EVI)* to understand savannas:

- ▶ **Long-term Mean:** Shows average vegetation density.
- ▶ **Linear Decadal Trends:** Identifies areas with permanent gains (greening) or losses (browning) in vegetation.
- ▶ **Seasonal Amplitude:** Measures the “peak-to-trough” variation (high in seasonal grasslands, low in stable forests).
- ▶ **Application:** Understanding movements of migratory ungulates like wildebeest.

Northern Tanzania (2000–2011)

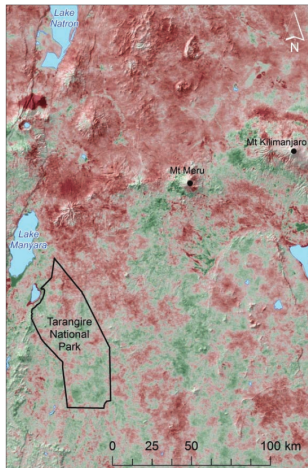
(a)



0 0.5

EVI mean (units)

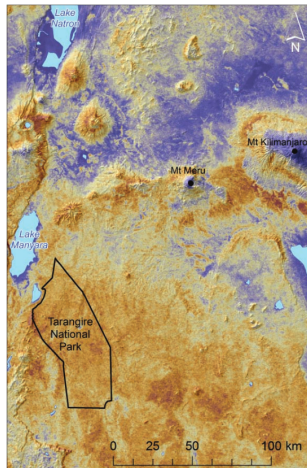
(b)



-0.1 +0.1

EVI trend (units per decade)

(c)



0 0.5

EVI seasonal amplitude (units)

Time series of MODIS EVI data. (a) Long-term mean. (b) Decadal trend. (c) Amplitude of seasonal cycle.

Modeling Spectral Response Patterns

Scientists use physical modeling to classify crops and estimate productivity.

- ▶ **Annual Crops:** Follow a **sigmoidal** time behavior.
- ▶ **Soil Background (G_0):** Generally remains constant.

The Sigmoidal Equation

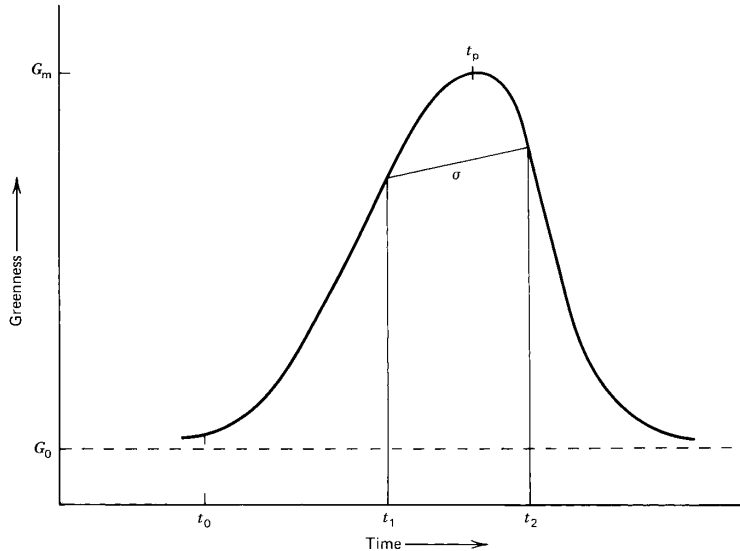
Greenness at time t is modeled using peak values and inflection points.

Dimensionality Reduction in Time Series

Instead of using 50+ raw images, we can characterize a season using three primary features:

1. G_m : **Peak Greenness** (The maximum intensity).
 2. t_p : **Time of Peak Greenness** (When it occurs).
 3. σ : **Width of the Profile** (Duration between inflection points t_1 and t_2).
- These three variables account for $> 95\%$ of the information in the original temporal data.

Temporal profile model for greenness



Key parameters include spectral emergence date (t_0), time (t_p) of peak greenness (G_m), and width of the profile (σ).

Conclusion: The Data Marathon

- ▶ Time series analysis transforms remote sensing from a “map-making” tool into a “process-monitoring” tool.
- ▶ **Agrophysical Mapping:** Variables like G_m and t_p relate directly to plant health and harvest timing.
- ▶ **Scale:** Essential for tracking global change, from desertification to the impact of abnormal flooding.

Homework

- ▶ Pick one of the following papers about the Continuous Change Detection (CCDC) algorithm:
 1. Zhu, Z., & Woodcock, C. E. (2014). Continuous change detection and classification of land cover using all available Landsat data. Remote sensing of Environment, 144, 152-171. <https://doi.org/10.1016/j.rse.2014.01.011>
 2. Arévalo, P., Bullock, E.L., Woodcock, C.E., Olofsson, P., (2020). A Suite of Tools for Continuous Land Change Monitoring in Google Earth Engine. Front. Clim. 2. <https://doi.org/10.3389/fclim.2020.576740>
- ▶ Prepare a reading report following the L^AT_EX template you will find in the Lab04 folder.
- ▶ Submit your report (one per student), via Brightspace™ assignment, by **21 days** from now, before the class on that day.